

UNIT 2: RENEWABLE RESOURCES

Class Time: Minimum 75 Minutes

Description

This lesson explores the use and misuse of the earth's precious renewable and non-renewable resources. Students begin the lesson by participating in a simulation game on renewable resources. Through the example of agriculture, students explore consumption patterns and the distribution of wealth in the world. Students identify the dangers of over-consuming and develop management techniques for ensuring the sustainability of resources (See Activity 5: Share the Wealth).

Now that students have identified the management techniques used in sustaining agricultural land, students explore exactly how finite the earth's fertile land really is. In a simulation game where the earth is represented by an apple, the earth is broken down into its components, all the way to the miniscule amount that is fertile soil. This demonstrates to the students how finite the earth really is and by comparing the total amount of fertile soil to population, students can understand the necessity of adopting sustainable practices. (See Activity 6: The Apple of our Eye). At this point, it is useful to discuss how your students scored on their ecological footprint quiz to see how much fertile soil their daily activities consume (See Activity 7: The Ecological Footprint). The unit concludes with a return to the yarn of life. This time, students return to the value they identified at the beginning of the module and then come up with one action they can take to reduce any threats to the sustainability of what they value (See Activity 8: The Yarn of Life Part 2).

Curriculum Expectations

Share the Wealth

Overall Expectations

HEV.01 – analyse selected global trends and evaluate their effects on people and environments at the local, national, and global level.

HEV.02 – analyse geographic issues that arise from the impact of human activities on the environment in different regions of the world.

Specific Expectations

HE1.01 – demonstrate an understanding of how human-induced changes in natural systems can diminish their capacity for supporting human activity.

HE1.03 – explain how people perceive resources and sustainable development differently at different times and in different places.

HE2.05 – evaluate some of the ways of promoting sustainable development and assess their effectiveness in selected places and regions of the world.

The Apple of Our Eye

Specific Expectations

HE1.03 – explain how people perceive resources and sustainable development differently at different times and in different places.

The Ecological Footprint

Specific Expectations

UC2.02 – evaluate the sustainability of selected trends related to consumption of the earth's resources.

Prior Knowledge and Skills

- Knowledge of global environmental trends (climate change, biodiversity, pollution, desertification, deforestation) and a familiarisation with the international agreements that address these trends (Kyoto protocol, the Convention on Biodiversity, etc).
- An understanding of human development trends and how they influence geographic issues (birth/death rates, GDP, child mortality, emissions, etc).
- Statistical Analysis, GIS basic knowledge.

Planning Notes

- Photocopy one copy of Worksheets 7 to 10 for each group for “Sharing the Wealth”.
- Bring a bag of beads, buttons or bean seeds for the “Share the Wealth” activity. To be safe, you need 400 beans.
- Bring an apple and paring knife to class.
- Remind students to bring their ecological footprint quiz to class.
- Photocopy The Ecological Footprint Explained (See Appendix 5)
- Review The Suggested Homework Exercises 2 and 3 and decide if you will assign them.
- Review the Glossary of terms and be prepared to define any of these terms for the class.

Teaching/Learning Strategies

Most of the learning strategies are described in each activity, but here are some of the overall strategies for the unit.

- Students will explore the use and misuse of the earth's precious renewable and non-renewable resources through a variety of simulation games.
- Through the example of agriculture, students identify in the activity ‘Share the Wealth’ the dangers of over-consuming. They develop management techniques to ensure the sustainability of resources.
- In the activity ‘The Apple of our Eye’, the students will explore how finite the earth's fertile land really is. They will explain how people perceive resources and sustainable development differently at different times and in different places.
- Students, with the help of ‘The Ecological Footprint’, will evaluate the sustainability of selected trends related to consumption.
- Students will discuss some of the connections between their values, how they interact, and which actions they will take to reduce any threats to the sustainability of what they value.
- Introduce the students to any new vocabulary and the concepts listed in each activity. Refer them to the glossary.
- Refer students to the accompanying material, the supplementary sheets, and where to go to find more information.
- Assign students the suggested homework and encourage students to analyse more in-depth the information provided and to research the websites given for this unit.
- Monitor the group work and the simulation games so students can gain maximum benefits from those activities.

Accommodations

Frequent monitoring and assistance should be provided. Accommodations that are normally provided in the regular classroom for students with special needs should be provided. Here are some strategies for helping them.

- Information should be presented to as many modalities as possible through oral presentations, board notes, diagrams, class discussions, activity-based learning.
- Instructions should be clear, concise, to the point, verbally rehearsed by the student, properly sequenced, reviewed after a time lapse, written down on the board or in a notebook.
- Assignments should be outlined in writing, written in the students' homework/assignment book, written (and left) on the blackboard, broken down by step with the teacher, rehearsed / modelled, and scheduled step by step.
- Questioning should be clear, concise, allowing for appropriate response time. Students need time to organise their thoughts, match them with the appropriate vocabulary, put them in correct syntax, sequence the ideas and finally express them with good speech. All these skills, most others take for granted.
- Encourage students to question for clarification and additional information. Learn to speak at a slower rate of speech. Avoid figurative language unless it has been specifically pre-taught. Encourage and evaluate oral responses and class participation. Allow extra time (e.g. homework) for students to complete assignments that others students may finish in class time. Create extra practice opportunities with new vocabulary or concepts. Provide overviews or lesson highlights at the beginning of class. Discuss and simplify sophisticated vocabulary or language. Finally, give several short assignments rather than one long one.